Code No: 135CH **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
FUNDAMENTALS OF MECHANICAL ENGINEERING
(Common to CE, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) State and explain zeroth law of thermodynamics and discuss its applications.
b) Explain the importance of specific heat capacity and specific enthalpy. [7+8]

2. Explain in detail the reasons for global warming and suggest some remedies to mitigate the effect of global warming. [15]

3. Explain the relation between Cp and Cv for perfect gas and discuss the salient points. [15]

4. Explain the functioning and accessories of Lancashire and Babcock and Wilcox boilers. [15]

5. Explain how do you estimate the thermal efficiency of Carnot cycle, otto cycle and diesel cycles. [15]

6. Explain the working principle of two- stroke petrol engine with a suitable diagram. [15]

7. Explain the working of vapor compression refrigeration system with the help of a neat sketch. [15]

8. Explain the construction and operational features of centrifugal clutches and discuss their applications. [15]

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